

TradingAgents V2 Benchmark — Comprehensive Results

Date: 2026-02-02 **Benchmark Version:** V2 (look-ahead free, no HOLD fallback) **Ticker:** AMBA (Ambarella Inc.) — "Hard" difficulty (25–42% accuracy in V1) **Period:** January – April 2025 (4 monthly snapshots) **Price Trajectory:** \$76.72 → \$47.99 (–37.5% over 4 months, high volatility bear market)

Conclusion

Claude Opus 4.5 single-agent was the best overall strategy, achieving the highest accuracy, highest P&L, lowest drawdown, best confidence calibration, and best 5-day forward returns. It outperformed both OpenAI gpt-5.1-codex-mini single-agent and both multi-agent configurations.

Multi-agent strategies are currently broken. Both OpenAI and Anthropic multi-agent setups produced 100% HOLD decisions with near-neutral confidence (0.45–0.55), generating zero P&L. This HOLD bias is architectural — caused by the risk manager prompt — not model-dependent. Swapping to more expensive models does not fix it. The multi-agent pipeline must not be used for live decisions until the risk manager prompt is reworked.

Single-agent strategies are 5–13x cheaper and more profitable than multi-agent alternatives, making them the only viable architecture in the current codebase.

Final Results

Strategy Comparison

Metric	OpenAI Single	Opus Single	OpenAI MA	Opus+Sonnet MA
Next-Day Accuracy	25.0% (1/4)	50.0% (2/4)	50.0% (2/4)	50.0% (2/4)
5-Day Accuracy	75.0% (3/4)	75.0% (3/4)	25.0% (1/4)	25.0% (1/4)
Total P&L	\$645.82	\$696.25	\$0.00	\$0.00
5-Day P&L	\$2,445.99	\$3,081.54	\$0.00	\$0.00
Max Drawdown	\$91.25	\$40.83	\$0.00	\$0.00
Avg Confidence	0.33	0.69	0.50	0.50
Decision Mix	75% SELL, 25% HOLD	50% SELL, 50% HOLD	100% HOLD	100% HOLD
Est. Cost	~\$0.32	~\$1.80	~\$2.00	~\$10.40
P&L per \$ Spent	\$2,018	\$387	\$0	\$0

Decision-by-Decision

Date	AMBA Close	Market (1d)	Market (5d)	OpenAI Single	Opus Single	OpenAI MA	Opus+Sonnet MA
Jan 31	\$76.72	−1.84%	−1.30%	HOLD (0.40)	HOLD (0.55)	HOLD (0.50)	HOLD (0.45)
Feb 28	\$61.43	−7.52%	−13.61%	SELL (0.30)	SELL (0.85)	HOLD (0.50)	HOLD (0.45)
Mar 31	\$50.33	+0.26%	−17.21%	SELL (0.30)	SELL (0.70)	HOLD (0.50)	HOLD (0.55)
Apr 30	\$47.99	+0.35%	+6.36%	SELL (0.30)	HOLD (0.65)	HOLD (0.50)	HOLD (0.55)

P&L by Trade

Date	OpenAI Single	Opus Single	OpenAI MA	Opus+Sonnet MA
Jan 31	\$0.00	\$0.00	\$0.00	\$0.00
Feb 28	+\$737.08	+\$737.08	\$0.00	\$0.00
Mar 31	−\$40.83	−\$40.83	\$0.00	\$0.00
Apr 30	−\$50.42	\$0.00	\$0.00	\$0.00
Total	\$645.82	\$696.25	\$0.00	\$0.00

Key Findings

1. Opus 4.5 single-agent was the best overall strategy. It tied for the highest next-day accuracy (50%), delivered the highest P&L (\$696.25), had the lowest drawdown (\$40.83), and produced the best 5-day forward P&L (\$3,081.54). Its confidence signal was also the most informative — the 0.85-confidence February SELL was the single most profitable call in the benchmark.

2. Multi-agent HOLD bias is model-independent. Both OpenAI and Anthropic multi-agent strategies produced identical 100% HOLD behavior with 0.45–0.55 confidence. This was also observed in V1 (503 tasks) where multi-agent HOLD rates ranged from 88–100% across 5 different multi-agent configurations. The root cause is the risk manager prompt, not the underlying model.

3. Single-agent strategies are more decisive and more profitable. Single-agent strategies took positions (50–75% SELL) while multi-agent was paralyzed (0% SELL). Even with some wrong calls, the net P&L was positive because winning trades compensated for losses.

4. Opus showed superior confidence calibration. Opus varied confidence from 0.55 to 0.85 (range: 0.30), expressing higher conviction on stronger signals. GPT-5.1-codex-mini varied only from 0.30 to 0.40 (range: 0.10), unable to differentiate signal strength. Opus's high-confidence calls (>0.6) were correct 67% of the time next-day.

5. LLMs predict weekly trends better than daily moves. Both single-agent strategies achieved 75% 5-day accuracy vs 25–50% next-day accuracy. Fundamental/news-driven analysis provides medium-term directional signals, not intraday precision.

6. The April divergence was the only meaningful difference between models. OpenAI and Opus single-agent agreed on 3 of 4 decisions. On April 30, Opus correctly held (recognizing the bottoming pattern) while OpenAI kept selling. That single call accounts for the entire \$50.42 P&L gap.

Recommendations

1. **Do not use multi-agent for live decisions** until the risk manager prompt is reworked
2. **Use single-agent Opus 4.5 as the default strategy** — best accuracy, best P&L, best calibration
3. **Consider single-agent OpenAI gpt-5.1-codex-mini as a budget alternative** — 5.6x cheaper, similar P&L, lower accuracy
4. **Evaluate on a 5-day holding period** rather than next-day — both models show much stronger signal at the weekly horizon
5. **Extend this benchmark to 50+ tasks** for statistical validity before drawing firm conclusions
6. **Fix the risk manager:** reduce its authority to override, add explicit anti-HOLD-bias instructions, or make it advisory-only

Limitations

- **4 data points per strategy** — too small for statistical significance; all findings are directional only
- **Single ticker (AMBA)** — performance could differ on other stock types
- **No BUY signals** — no strategy issued a BUY across all 16 decisions
- **Cost estimates are approximate** — token usage was not tracked; costs based on per-task averages
- **Anthropic MA used cached analyst reports** from the single-agent run (same inputs, different decision pipeline)
- **OpenAI run was stopped at 8/96 tasks** — only AMBA Jan–Apr covered

Detailed Analysis

Everything below provides the supporting evidence, methodology, and raw data behind the conclusions above.

1. Benchmark Design

Objective

Compare OpenAI (gpt-5.1-codex-mini) and Anthropic (claude-opus-4-5, claude-sonnet-4-5) across both single-agent and multi-agent architectures on the same data points, using the V2 evaluation framework which eliminates look-ahead bias and forces explicit BUY/SELL/HOLD decisions.

Ticker Selection

AMBA was selected as a "hard" ticker based on V1 benchmark data (503 tasks across 6 tickers, 7 strategies):

Ticker	V1 Best Accuracy	Accuracy Range	Difficulty	Rationale
RAPT	17%	8–17%	Hardest	Small-cap biotech (\$3), sparse news, extreme volatility

Ticker	V1 Best Accuracy	Accuracy Range	Difficulty	Rationale
SNOW	42%	8–42%	Hard	Cloud software, wide strategy variance
AMBA	42%	25–42%	Hard	Mid-cap semiconductor, earnings-driven volatility
ZTS	58%	50–58%	Moderate	Large-cap animal health, stable
ET	67%	50–67%	Easy	Energy MLP, consistent price behavior
SUPN	75%	50–75%	Easiest	Mid-cap pharma, clearest signals

AMBA tests models' ability to handle earnings-driven semiconductor volatility with limited, ambiguous signals.

Strategy Configurations

Four strategies were tested, each on the same 4 analysis dates:

Strategy Label	Architecture	Provider	Deep Model	Quick Model
ma_5.1-codex-mini	Multi-agent graph	OpenAI	gpt-5.1-codex-mini	gpt-5.1-codex-mini
single_5.1-codex-mini	Single-agent baseline	OpenAI	gpt-5.1-codex-mini	—
single_opus-4-5	Single-agent baseline	Anthropic	claude-opus-4-5-20251101	—
ma_opus-4-5+sonnet-4-5	Multi-agent graph	Anthropic	claude-opus-4-5-20251101	claude-sonnet-4-5-20250929

Multi-Agent Graph Architecture

The multi-agent pipeline runs through:

1. **4 Analyst nodes** (Sonnet/codex-mini): Market, Fundamentals, News, Social Media
2. **2 Researcher nodes** (Sonnet/codex-mini): Bull Researcher, Bear Researcher
3. **Research Manager** (Opus/codex-mini): Synthesizes bull/bear arguments
4. **Risk Manager** (Opus/codex-mini): Evaluates and challenges the recommendation
5. **Trader** (Opus/codex-mini): Makes final BUY/SELL/HOLD decision

The single-agent baseline feeds all raw data (market, fundamentals, news, sentiment) into a single LLM call that produces a decision directly.

Common Configuration

All runs shared identical non-LLM settings:

Setting	Value
Data Vendors — Core Stock	Polygon
Data Vendors — Technical Indicators	Polygon
Data Vendors — Fundamentals	Alpha Vantage

Setting	Value
Data Vendors — News	Polygon + Alpha Vantage
Data Vendors — Social Sentiment	StockTwits
News Limits (default / 3mo / 6mo / 12mo)	60 / 200 / 500 / 600
Sentiment Limits (default / 3mo / 6mo / 12mo)	50 / 200 / 500 / 500
News Monthly Bucketing	Enabled
News Company Fallback	Enabled
Max Debate Rounds	1
Max Risk Discussion Rounds	1
Memory System	Disabled
ACE Framework	Disabled
Commission per Trade	\$15.00
Simulated Portfolio	\$10,000

2. V1 → V2 Methodology Changes

V2 introduced critical fixes that make results non-comparable with V1:

Issue	V1 Behavior	V2 Fix
Look-ahead bias	<code>get_current_quote()</code> fetched live prices	Uses <code>decision_date_close</code> from historical OHLCV only
HOLD fallback	Parse failures silently defaulted to HOLD 0.5	Raises <code>ValueError</code> — forces real model decision
Content normalization	Raw <code>AIMessage</code> objects passed through pipeline	<code>normalize_content()</code> applied at all agent boundaries
Type hints	Mixed <code>ChatOpenAI</code> types	<code>BaseChatModel</code> throughout for provider-agnostic code
Anthropic instantiation	<code>base_url</code> (OpenAI param) passed to <code>ChatAnthropic</code>	Auto-discovery from <code>ANTHROPIC_API_KEY</code> ; <code>max_tokens=8192</code>
5-day validation	Not tracked	<code>day5_change_pct</code> and <code>day5_correct</code> fields added
Evaluation framework	Manual runs, no checkpointing	<code>evaluate_v2.py</code> with checkpoints, resume, compare, dry-run

3. Run Inventory

Run ID	Provider	Strategy	Tasks	Completed	Failed	Status
v2_20260202_153754	OpenAI	ma + single (gpt-5.1-codex-mini)	96 total (8 AMBA)	8	0	Stopped early
v2_20260202_161514	Anthropic	single (claude-opus-4-5)	4	4	0	Complete
v2_20260202_162458	Anthropic	multi-agent (opus+sonnet)	4	4	0	Complete

Note: The OpenAI run was configured for 96 tasks (4 tickers × 12 months × 2 strategies) but was stopped after 8 tasks covering only AMBA Jan–Apr 2025. The remaining 88 tasks for RAPT, ET, ZTS are pending and can be resumed with `--resume v2_20260202_153754`.

4. Market Context — AMBA Jan–Apr 2025

AMBA (Ambarella Inc.) experienced a sustained decline through Q1 2025:

Date	Close Price	Next-Day Change	5-Day Change	Regime	Volatility
2025-01-31	\$76.72	-1.84%	-1.30%	Sideways	High
2025-02-28	\$61.43	-7.52%	-13.61%	Bear	High
2025-03-31	\$50.33	+0.26%	-17.21%	Bear	High
2025-04-30	\$47.99	+0.35%	+6.36%	Bear	High

Price trajectory: \$76.72 → \$47.99 (−37.5% over 4 months)

Key observations:

- **January:** Sideways chop, ambiguous signals. Next day slightly down, week slightly down.
- **February:** Sharp bear break. Next day crashed −7.5%, week dropped −13.6%. The clearest sell signal.
- **March:** Continued decline but next day was flat (+0.26%). The 5-day continued down −17.2%. A trap — the daily signal said "flat" but the weekly trend was deeply bearish.
- **April:** Bottoming. Next day +0.35%, week recovered +6.4%. The first bullish reversal signal.

The "correct" strategy in hindsight was: SELL Feb, SELL Mar (for 5-day), HOLD or BUY Apr.

5. Trade-by-Trade Detail

5.1 OpenAI Multi-Agent (ma_5.1-codex-mini)

#	Date	Decision	Conf.	Entry	Next Close	Next-Day Δ	5-Day Δ	P&L	Correct?	5d Correct?
1	2025-01-31	HOLD	0.50	\$76.72	\$75.31	-1.84%	-1.30%	\$0.00	N	Y
2	2025-02-28	HOLD	0.50	\$61.43	\$56.81	-7.52%	-13.61%	\$0.00	N	N

#	Date	Decision	Conf.	Entry	Next Close	Next-Day Δ	5-Day Δ	P&L	Correct?	5d Correct?
3	2025-03-31	HOLD	0.50	\$50.33	\$50.46	+0.26%	-17.21%	\$0.00	Y	N
4	2025-04-30	HOLD	0.50	\$47.99	\$48.16	+0.35%	+6.36%	\$0.00	Y	N

Totals: 4/4 HOLD, 2/4 correct (50.0%), \$0.00 P&L, 1/4 5-day correct (25.0%)

5.2 OpenAI Single-Agent ([single_5.1-codex-mini](#))

#	Date	Decision	Conf.	Entry	Next Close	Next-Day Δ	5-Day Δ	P&L	Correct?	5d Correct?
1	2025-01-31	HOLD	0.40	\$76.72	\$75.31	-1.84%	-1.30%	\$0.00	N	Y
2	2025-02-28	SELL	0.30	\$61.43	\$56.81	-7.52%	-13.61%	+\$737.08	Y	Y
3	2025-03-31	SELL	0.30	\$50.33	\$50.46	+0.26%	-17.21%	-\$40.83	N	Y
4	2025-04-30	SELL	0.30	\$47.99	\$48.16	+0.35%	+6.36%	-\$50.42	N	N

Totals: 3 SELL + 1 HOLD, 1/4 correct (25.0%), +\$645.82 P&L, 3/4 5-day correct (75.0%)

Notes:

- Feb SELL was the big winner: \$737 profit on a -7.5% next-day crash
- Mar SELL lost \$41 (market was flat next day, though 5-day confirmed the sell)
- Apr SELL lost \$50 (market bounced, wrong direction)
- All decisions had low confidence (0.30–0.40), suggesting the model was not sure but biased toward action

5.3 Anthropic Single-Agent ([single_opus-4-5](#))

#	Date	Decision	Conf.	Entry	Next Close	Next-Day Δ	5-Day Δ	P&L	Correct?	5d Correct?
1	2025-01-31	HOLD	0.55	\$76.72	\$75.31	-1.84%	-1.30%	\$0.00	N	Y
2	2025-02-28	SELL	0.85	\$61.43	\$56.81	-7.52%	-13.61%	+\$737.08	Y	Y
3	2025-03-31	SELL	0.70	\$50.33	\$50.46	+0.26%	-17.21%	-\$40.83	N	Y

#	Date	Decision	Conf.	Entry	Next Close	Next-Day Δ	5-Day Δ	P&L	Correct?	5d Correct?
4	2025-04-30	HOLD	0.65	\$47.99	\$48.16	+0.35%	+6.36%	\$0.00	Y	N

Totals: 2 SELL + 2 HOLD, 2/4 correct (50.0%), +\$696.25 P&L, 3/4 5-day correct (75.0%)

Notes:

- Feb SELL matched OpenAI's call but with **much higher confidence** (0.85 vs 0.30)
- Mar SELL also matched, but with higher confidence (0.70 vs 0.30) — still wrong next-day
- Apr was the key differentiator: Opus **correctly held** (0.65 confidence) while OpenAI single sold. This saved \$50.42 and turned a wrong call into a right one.
- Overall confidence much higher (avg 0.69 vs 0.33), indicating Opus expressed more conviction in its analysis

5.4 Anthropic Multi-Agent (ma_opus-4-5+sonnet-4-5)

#	Date	Decision	Conf.	Entry	Next Close	Next-Day Δ	5-Day Δ	P&L	Correct?	5d Correct?
1	2025-01-31	HOLD	0.45	\$76.72	\$75.31	-1.84%	-1.30%	\$0.00	N	Y
2	2025-02-28	HOLD	0.45	\$61.43	\$56.81	-7.52%	-13.61%	\$0.00	N	N
3	2025-03-31	HOLD	0.55	\$50.33	\$50.46	+0.26%	-17.21%	\$0.00	Y	N
4	2025-04-30	HOLD	0.55	\$47.99	\$48.16	+0.35%	+6.36%	\$0.00	Y	N

Totals: 4/4 HOLD, 2/4 correct (50.0%), \$0.00 P&L, 1/4 5-day correct (25.0%)

Notes:

- Identical pattern to OpenAI multi-agent: 100% HOLD, 50% next-day accuracy
- Confidence range was narrow (0.45–0.55), all hovering around 0.5
- Critically missed the Feb –7.5% crash — the strongest signal in the dataset
- The multi-agent graph's risk manager overrides actionable signals with HOLD regardless of LLM provider

6. Pivotal Decisions

February 28 — The Clear Signal

AMBA closed at \$61.43, already down 20% from January. Next day crashed another –7.5%, 5-day dropped –13.6%.

- **Opus single** called SELL with 0.85 confidence — the highest conviction of any call in the entire benchmark

- **OpenAI single** also called SELL but with only 0.30 confidence — barely above "uncertain"
- **Both multi-agent strategies missed this entirely** — held at 0.45–0.50 confidence

This was the clearest bearish signal in the dataset and the largest single-trade profit opportunity (+\$737). The fact that both multi-agent strategies held through a –7.5% crash demonstrates the severity of the HOLD bias.

April 30 — The Divergence

AMBA had bottomed at \$47.99 after 4 months of decline. Next day +0.35%, 5-day rebounded +6.4%.

- **Opus single correctly held** (0.65 confidence) — recognized the bottoming pattern
- **OpenAI single kept selling** (0.30 confidence) — continued the bearish momentum bias
- This single decision accounted for the entire P&L gap between the two single-agent strategies (\$696.25 vs \$645.82)

7. Decision Agreement Matrix

How often each pair of strategies agreed on the same decision:

	OpenAI MA	OpenAI Single	Opus Single	Opus+Sonnet MA
OpenAI MA	—	1/4 (25%)	2/4 (50%)	4/4 (100%)
OpenAI Single	1/4	—	3/4 (75%)	1/4 (25%)
Opus Single	2/4	3/4	—	2/4 (50%)
Opus+Sonnet MA	4/4	1/4	2/4	—

- The two multi-agent strategies produced **identical decisions** (100% agreement) — both all-HOLD
- The two single-agent strategies agreed on **3 of 4 decisions** — diverged only on April
- Multi-agent and single-agent have low agreement (25–50%), confirming they operate very differently

8. Confidence Analysis

Distribution by Strategy

Strategy	Min	Max	Avg	Std Dev	Range
OpenAI MA	0.50	0.50	0.50	0.00	0.00
OpenAI Single	0.30	0.40	0.33	0.05	0.10
Opus Single	0.55	0.85	0.69	0.13	0.30
Opus+Sonnet MA	0.45	0.55	0.50	0.05	0.10

Opus Single: Confidence vs Correctness

Confidence	Decision	Next-Day Correct	5-Day Correct
0.85	SELL (Feb)	Y	Y
0.70	SELL (Mar)	N	Y

Confidence	Decision	Next-Day Correct	5-Day Correct
0.65	HOLD (Apr)	Y	N
0.55	HOLD (Jan)	N	Y

High-confidence calls (>0.6): 2/3 correct next-day (67%), 2/3 correct 5-day (67%) Low-confidence calls (≤0.6): 0/1 correct next-day (0%), 1/1 correct 5-day (100%)

Opus expressed higher confidence on its SELL calls and was right on the strongest one (Feb). The confidence signal contains useful information — high-confidence Opus calls were more likely to be correct.

Multi-Agent Confidence Collapse

Both multi-agent strategies produced near-constant confidence (0.45–0.55), essentially defaulting to "uncertain." The multi-agent debate process converges to the mean rather than producing differentiated conviction. The risk manager's challenge step suppresses confidence regardless of the underlying model.

9. 5-Day Forward Accuracy

Strategy	Next-Day Acc.	5-Day Acc.	5-Day P&L	Next-Day P&L
OpenAI MA	50.0%	25.0%	\$0.00	\$0.00
OpenAI Single	25.0%	75.0%	\$2,445.99	\$645.82
Opus Single	50.0%	75.0%	\$3,081.54	\$696.25
Opus+Sonnet MA	50.0%	25.0%	\$0.00	\$0.00

Both single-agent strategies achieved 75% 5-day accuracy vs 25–50% next-day accuracy. This suggests LLM-based analysis is better suited to medium-term directional calls than daily price prediction, which aligns with the fundamental/news-driven nature of the input data.

The 5-day P&L gap is dramatic: Opus single would have generated \$3,081 over 4 trades on a 5-day basis, while multi-agent strategies still generated \$0 because they never took a position.

10. Cost Efficiency Analysis

Estimated Cost per Strategy

Strategy	Tasks	Est. \$/Task	Total Cost	Accuracy	P&L	P&L/\$ Spent
OpenAI Single	4	\$0.08	~\$0.32	25%	\$645.82	\$2,018
Opus Single	4	\$0.45	~\$1.80	50%	\$696.25	\$387
OpenAI MA	4	\$0.50	~\$2.00	50%	\$0.00	\$0
Opus+Sonnet MA	4	\$2.60	~\$10.40	50%	\$0.00	\$0

Cost per Correct Decision

Strategy	Correct	Total Cost	\$/Correct
OpenAI Single	1	~\$0.32	~\$0.32
Opus Single	2	~\$1.80	~\$0.90
OpenAI MA	2	~\$2.00	~\$1.00
Opus+Sonnet MA	2	~\$10.40	~\$5.20

Observations:

1. OpenAI single was cheapest (~\$0.32 total) with the best P&L-per-dollar — but skewed by one lucky SELL

2. Opus single had the best accuracy-to-cost ratio for a moderate cost (\$1.80)

3. Multi-agent Anthropic was the most expensive (~\$10.40) for zero P&L — worst value by any metric

4. Multi-agent adds ~5–13x cost over single-agent for the same accuracy and less P&L

11. Multi-Agent HOLD Bias Analysis

The Pattern

Both multi-agent strategies produced **identical 100% HOLD behavior**:

Date	OpenAI MA	Anthropic MA	Match?
Jan 31	HOLD (0.50)	HOLD (0.45)	Yes
Feb 28	HOLD (0.50)	HOLD (0.45)	Yes
Mar 31	HOLD (0.50)	HOLD (0.55)	Yes
Apr 30	HOLD (0.50)	HOLD (0.55)	Yes

100% agreement. 100% HOLD.

V1 Corroboration (503 tasks)

The V1 benchmark showed the same pattern at scale:

V1 Strategy	HOLD Rate	Accuracy	P&L
ma_5.2 (gpt-5.2)	100%	40.3%	\$0.00
ma_5.2+5.1-codex-mini	96%	38.0%	\$26.32
ma_4o (gpt-4o)	94%	43.1%	-\$408.82
ma_5.1-codex-mini	88%	33.3%	-\$389.58
single_5.1-codex-mini	71%	45.8%	\$1,173.32
single_5.2	88%	40.3%	\$519.09
single_4o	90%	40.3%	-\$759.35

Multi-agent strategies had 88–100% HOLD rates across 72 tasks each in V1.

Root Cause

The HOLD bias is **architectural, not model-dependent**:

1. It persists across OpenAI (gpt-5.1-codex-mini) and Anthropic (opus-4-5 + sonnet-4-5)
2. It persists across model generations (V1: gpt-4o, gpt-5.2; V2: gpt-5.1-codex-mini, opus)
3. The confidence collapse to 0.45–0.55 is identical regardless of provider

The culprit is the **risk manager agent prompt** ([tradingagents/agents/managers/risk_manager.py](#)), which is designed to challenge and moderate the recommendation. When combined with the debate structure (bull vs bear researchers), the system converges to "insufficient evidence" → HOLD at the neutral confidence midpoint.

12. Comparison with V1 Benchmark (503 tasks)

V1 Context

The V1 benchmark (run ID: [bt_20260129_173611](#)) ran 503 tasks across 6 tickers, 12 months, and 7 strategies. It used the old evaluation framework which had look-ahead bias in price fetching and allowed HOLD fallbacks on parse failures.

AMBA Specifically — V1 vs V2

V1 AMBA results (12 months, full year):

V1 Strategy	AMBA Accuracy	AMBA P&L
ma_5.2+5.1-codex-mini	42%	\$169
ma_4o	42%	\$99
ma_5.2	33%	\$0
single_5.1-codex-mini	33%	\$445
ma_5.1-codex-mini	25%	-\$21
single_5.2	25%	-\$355
single_4o	25%	-\$739

V2 AMBA results (4 months, Jan–Apr 2025):

V2 Strategy	AMBA Accuracy	AMBA P&L
Opus Single	50%	\$696
OpenAI MA	50%	\$0
Opus+Sonnet MA	50%	\$0
OpenAI Single	25%	\$646

Caution: V1 and V2 results are NOT directly comparable due to:

- V1 had look-ahead bias (inflated accuracy for some decisions)
- V2 covers only 4 of 12 months
- V2 removed HOLD fallback (some V1 "decisions" were parse failures silently counted as HOLD)
- Different data vendor configurations

After removing methodological flaws, AMBA remains a 25–50% accuracy ticker, consistent with V1's 25–42% range.

13. Raw Run References

Run Files

Run	Checkpoint	Outcomes	Report
OpenAI (v2_20260202_153754)	checkpoint.json	outcomes.json	report.md
Opus Single (v2_20260202_161514)	checkpoint.json	outcomes.json	report.md
Opus+Sonnet MA (v2_20260202_162458)	checkpoint.json	outcomes.json	report.md

Pairwise Comparison Reports

Comparison	File
OpenAI vs Opus Single	V2_COMPARE_...153754_vs_...161514.md
OpenAI vs Opus+Sonnet MA	V2_COMPARE_...153754_vs_...162458.md
Opus Single vs Opus+Sonnet MA	V2_COMPARE_...161514_vs_...162458.md

Resume / Extend Commands

```
# Resume the OpenAI run (88 remaining tasks: RAPT, ET, ZTS)
python -m cli.evaluate_v2 --resume v2_20260202_153754 --wait-on-rate-limit

# Run Anthropic single on more tickers (e.g., RAPT)
python -m cli.evaluate_v2 --single-only \
    --tickers RAPT --start 2025-01-28 --end 2025-05-28 \
    --provider anthropic \
    --deep-model claude-opus-4-5-20251101 \
    --quick-model claude-opus-4-5-20251101 \
    --wait-on-rate-limit

# Compare any two runs
python -m cli.evaluate_v2 --compare <run_id_a> <run_id_b>

# Dry-run cost estimate before committing
python -m cli.evaluate_v2 --dry-run --tickers AMBA,RAPT \
    --provider anthropic --deep-model claude-opus-4-5-20251101 \
    --quick-model claude-opus-4-5-20251101
```

V1 Benchmark Reference

The V1 benchmark report (503 tasks, 6 tickers, 7 strategies, 12 months) is at: [BENCHMARK_REPORT_2026-02-01_RAPT_SUPN_ZTS.md](#)